Contract Risk Analysis via Context-Conditioned LLM Adaptation

Aarav Singh Luthra, Devyansh Choudhary, Saksham Bali, Uday Sodhi

Abstract

Legal contracts such as lease agreements, employment contracts, freelance agreements, and digital terms and conditions often contain clauses that impose financial or legal obligations that may not be immediately clear to non-expert users. In this project, we develop a contract analysis system that extends a pretrained instruction-tuned Large Language Model (LLM) with structured, clause-level risk extraction capabilities. The system first preprocesses uploaded contracts through deterministic document parsing and clause segmentation to produce semantically coherent clause units. It then performs structured risk scoring conditioned on user role, contract domain, and risk tolerance using parameter-efficient fine-tuning via Low-Rank Adaptation (LoRA). The model generates categorical risk levels, calibrated risk scores, and concise natural-language explanations tailored to the specified user context. If time permits, we further integrate a Retrieval-Augmented Generation (RAG) module to support grounded cross-questioning over uploaded contracts. The system is evaluated using structured output accuracy, calibration analysis, contextual sensitivity measurements, and human evaluation of generated explanations.

1 Introduction

Legal contracts are complex textual artifacts that encode rights, obligations, liabilities, and procedural constraints. Individuals frequently engage with employment agreements, lease contracts, freelance service agreements, non-disclosure agreements, and digital terms and conditions. Interpreting clause-level implications in such documents requires careful attention to language that may signal risk depending on the reader's role and priorities.

Instruction-tuned Large Language Models (LLMs) provide strong general-purpose reasoning capabilities and can generate natural-language explanations for textual inputs. However, out-of-the-box models are not explicitly trained to produce structured, clause-level risk assessments conditioned on user role and contextual attributes.

In this project, we aim to adapt a pretrained instruction-tuned LLM to perform structured, context-conditioned legal risk extraction. The system:

1. Preprocesses uploaded contracts and segments them into coherent clauses.
2. Performs clause-level risk scoring conditioned on user role, contract domain, and risk tolerance.
3. Produces structured outputs containing risk level, numerical score, and concise explanation.
4. Optionally supports grounded cross-questioning via retrieval augmentation.

Our objective is to study how parameter-efficient adaptation and explicit context conditioning can enable a pretrained language model to internalize user-aware legal risk interpretation in a controlled and measurable manner.

2 Problem Formulation

Let a contract be represented as an ordered set of clauses:

$$C = \{c_1, c_2, \ldots, c_n\}$$

where each clause c_i is a semantically coherent unit extracted from a document belonging to domain D (e.g., Lease, Employment, Freelance, NDA, Terms & Conditions).

2.1 Preprocessing and Clause Segmentation

Before modeling, uploaded documents undergo a preprocessing pipeline:

5. Document parsing (PDF/DOCX → raw text).
6. Text normalization (removal of headers, footers, page numbers, and formatting artifacts).
7. Deterministic clause boundary detection using rule-based heuristics such as: numbered headings (e.g., "1.", "2.1"), section indicators ("Section", "Article"), and capitalized or stylized headers.
8. Sentence-level coherence checks to merge fragmented segments and ensure semantically meaningful clause units.

This pipeline produces stable and reproducible clause-level inputs for modeling.

Let U denote structured user context consisting of: User Role (Tenant, Employer, Employee, Freelancer, Consumer) and Risk Tolerance (Low, Medium, High). The core modeling objective is to learn:

$$f(c_i, U, D) \rightarrow (r_i, s_i, e_i)$$

where r_i is a categorical risk level (Low, Medium, High, Critical), $s_i \in [0,1]$ is a continuous risk score, and e_i is a natural-language explanation tailored to user context.

For interactive querying, we define:

$$g(C, U, Q) \rightarrow A$$

where Q is a user question and A is a grounded answer generated using retrieved clauses from C.

The system consists of two logical modules sharing a fine-tuned LLM core: (1) Risk Flagging Module—structured clause-level risk extraction, and (2) Cross-Questioning Module—retrieval-conditioned generation.

3 Related Work

3.1 Legal NLP

Legal NLP research has explored domain-adaptive pretraining, clause classification, and document-level inference. LegalBERT (Chalkidis et al., 2020) demonstrates benefits of legal-domain pretraining. The CUAD dataset (Hendrycks et al., 2021) provides clause-level annotations for commercial contracts. ContractNLI (Koreeda and Manning, 2021) frames contract reasoning as a natural language inference problem. These works provide datasets and modeling strategies relevant to legal text processing but do not explicitly study user-conditioned clause-level risk scoring.

3.2 Instruction-Tuned LLMs and Parameter-Efficient Adaptation

Instruction-tuned models such as LLaMA (Touvron et al., 2023) and Mistral (Jiang et al., 2023) demonstrate strong zero-shot reasoning capabilities. Low-Rank Adaptation (LoRA) (Hu et al., 2022) enables efficient fine-tuning by inserting low-rank adapter matrices into transformer layers while freezing base parameters. This approach allows domain-specific adaptation without full retraining.

3.3 Retrieval-Augmented Generation

Retrieval-Augmented Generation (Lewis et al., 2020) combines dense retrieval with language model generation to improve grounding and factual consistency. This framework can be applied to contract analysis to support clause-anchored question answering.

4 Methodology

4.1 Dataset Construction and Annotation

We construct a clause-level dataset from publicly available contracts across multiple domains: lease agreements, employment contracts, freelance agreements, NDAs, and digital Terms & Conditions. Each contract is processed through the preprocessing and clause segmentation pipeline described earlier. The resulting clause units form the annotation pool.

Each training example consists of: Domain, User Role, Risk Tolerance, Clause Text, Risk Level, Risk Score, and Natural-language Explanation.

Annotation Strategy

Given the cost of large-scale expert annotation, we adopt a structured semi-automatic pipeline:

9. A strong external LLM generates initial structured risk outputs.
10. Human annotators review and correct these outputs for consistency and accuracy.
11. Post-processing ensures balanced label distribution across risk categories.

Special attention is given to annotation consistency across user roles to ensure that role-conditioned

differences are meaningful and systematically represented. We target approximately 800–1500 clause-role examples for LoRA fine-tuning of a 7–8B parameter model. Dataset quality and internal consistency are central to this project, as the model's ability to learn context-conditioned behavior depends directly on supervision quality.

4.2 Risk Schema

We define four categorical risk levels: Low, Medium, High, and Critical. Risk Score $s_i \in [0,1]$ may be predicted directly or mapped from categorical outputs via calibrated thresholds. Optional risk type tags (Financial, Termination, Liability, Ambiguity) are used for interpretability but are not core training targets.

4.3 Base Model and LoRA Fine-Tuning

We use a pretrained instruction-tuned LLM (7–8B parameters) as the base model. Fine-tuning is performed using QLoRA with 4-bit quantization for memory efficiency, frozen base weights, and trainable low-rank adapters inserted into attention projection layers.

Training examples follow a structured instruction format:

```
User Role: [role] | Domain: [domain] | Risk Tolerance: [tolerance]

Clause: [text]
Output:
Risk Level: [label]
Risk Score: [score]
Explanation: [text]
```

Cross-entropy loss is computed over output tokens only. This enables the model to learn structured, context-aware risk extraction without architectural modification.

4.4 Context Conditioning

User role and domain are injected explicitly during both training and inference. This encourages learning of the joint mapping: Risk = f(Clause, Role, Domain). Contextual sensitivity is evaluated by presenting identical clauses under different user roles and measuring variation in predicted outputs.

4.5 Risk Flagging Module

For inference: (1) the user uploads a contract; (2) the contract is segmented into clauses; (3) each clause is passed to the fine-tuned LLM with user context; (4) structured outputs are parsed and stored; (5) clauses are ranked by predicted risk.

4.6 Cross-Questioning Module (Optional RAG)

If implemented: clause embeddings are computed using a pretrained sentence encoder and stored in a FAISS index. For each user query, top-k clauses are retrieved. Retrieved clauses and user context are passed to the LLM to generate a grounded response.

5 Evaluation Metrics

Structured Risk Prediction

Categorical risk level outputs are parsed and evaluated using Accuracy, Precision, Recall, and Macro F1-score.

Risk Score Calibration

If continuous scores are predicted, evaluation includes Mean Absolute Error (MAE) and Spearman correlation with annotated scores.

Contextual Sensitivity

For identical clauses evaluated under different roles: variance in predicted risk scores and comparison between role-conditioned and role-agnostic models.

RAG Evaluation

Grounding rate (overlap with retrieved clauses) and hallucination rate (human evaluation).

Human Evaluation

100 clause explanations are evaluated on Clarity, Role Relevance, and Perceived Correctness using a 1–5 Likert scale.

Limitations

This document describes a proposed system and experimental design. Evaluation results, ablation studies, and cross-domain generalization analyses are deferred to the full experimental report.

References

Ilias Chalkidis, Manos Fergadiotis, Prodromos Malakasiotis, Nikolaos Aletras, and Ion

Androutsopoulos. 2020. LEGAL-BERT: The muppets straight out of law school. In Findings of EMNLP.

Dan Hendrycks, Collin Burns, Anya Chen, and Spencer Ball. 2021. CUAD: An expert-annotated NLP dataset for legal contract review. In NeurIPS Datasets and Benchmarks.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In ICLR.

Yuta Koreeda and Christopher Manning. 2021. ContractNLI: A dataset for document-level natural language inference for contracts. In Findings of EMNLP.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, et al. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In NeurIPS.

Hugo Touvron, Thibaut Lavril, Gautier Izacard, et al. 2023. LLaMA: Open and efficient foundation language models. arXiv:2302.13971.